

```
!pip install lightkurve matplotlib numpy
```

```
Collecting lightkurve
```

```
  Downloading lightkurve-2.6.0-py3-none-any.whl.metadata (6.5 kB)
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```
Requirement already satisfied: matplotlib in /usr/local/lib/python3.13/c
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Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-p
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Requirement already satisfied: astropy>=5.0 in /usr/local/lib/python3.13
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Collecting astroquery>=0.3.10 (from lightkurve)
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  Downloading astroquery-0.4.11-py3-none-any.whl.metadata (6.5 kB)
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Requirement already satisfied: beautifulsoup4>=4.6.0 in /usr/local/lib/p
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Requirement already satisfied: bokeh>=2.3.2 in /usr/local/lib/python3.13
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Collecting fbpcas>=1.0 (from lightkurve)
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  Downloading fbpcas-1.0.tar.gz (11 kB)
```

```
  Preparing metadata (setup.py) ... done
```

```
Collecting memoization>=0.3.1 (from lightkurve)
```

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  Downloading memoization-0.4.0.tar.gz (41 kB)
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41.2/41.2 kB 1.1 MB
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  Preparing metadata (setup.py) ... done
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Requirement already satisfied: pandas<3.0.0,>=1.3.6 in /usr/local/lib/py
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Requirement already satisfied: patsy>=0.5.0 in /usr/local/lib/python3.13
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Requirement already satisfied: requests>=2.22.0 in /usr/local/lib/pythor
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```
Collecting s3fs>=2024.6.1 (from lightkurve)
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  Downloading s3fs-2026.7.0-py3-none-any.whl.metadata (1.2 kB)
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Requirement already satisfied: scikit-learn>=0.24.0 in /usr/local/lib/py
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Requirement already satisfied: scipy>=1.7 in /usr/local/lib/python3.13/c
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Requirement already satisfied: tqdm>=4.25.0 in /usr/local/lib/python3.13
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Collecting uncertainties>=3.1.4 (from lightkurve)
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  Downloading uncertainties-3.2.3-py3-none-any.whl.metadata (7.0 kB)
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Requirement already satisfied: urllib3>=1.23 in /usr/local/lib/python3.1
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Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/pythor
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Requirement already satisfied: cyclical>=0.10 in /usr/local/lib/python3.13
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Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/pytho
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Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/pytho
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Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3
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Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.13/di
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Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/pythor
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Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/py
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Requirement already satisfied: astropy-iers-data>=0.2026.6.22.1.23.34 in
```

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Requirement already satisfied: pyerfa>=2.0.1.1 in /usr/local/lib/python3
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Requirement already satisfied: PyYAML>=6.0.0 in /usr/local/lib/python3.1
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Requirement already satisfied: html5lib>=0.999 in /usr/local/lib/python3
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Requirement already satisfied: keyring>=15.0 in /usr/local/lib/python3.1
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Collecting pyvo>=1.5 (from astroquery>=0.3.10->lightkurve)
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  Downloading pyvo-1.9.1-py3-none-any.whl.metadata (4.2 kB)
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Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.1
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Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/li
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Requirement already satisfied: Jinja2>=2.9 in /usr/local/lib/python3.13/
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Requirement already satisfied: narwhals>=1.13 in /usr/local/lib/python3.
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Requirement already satisfied: tornado>=6.2 in /usr/local/lib/python3.13
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Requirement already satisfied: xyzservices>=2021.09.1 in /usr/local/lib/
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Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13
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Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dis
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Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/li
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/pyth
Collecting aiobotocore<4.0.0,>=2.19.0 (from s3fs>=2024.6.1->lightkurve)
  Downloading aiobotocore-3.9.1-py3-none-any.whl.metadata (31 kB)
Collecting fsspec<2026.7.1,>=2026.7.0 (from s3fs>=2024.6.1->lightkurve)
```

```
import lightkurve as lk
import matplotlib.pyplot as plt
import numpy as np

# 1. NASA Kepler Archive se data search aur download
target_star = "Kepler-10"
print(f"Searching data for {target_star}...")

search_result = lk.search_lightcurve(target_star, author="Kepler", quarter=5)
lc = search_result.download()

# 2. Pre-processing: Outliers aur noise hatana
lc_clean = lc.remove_nans().remove_outliers(sigma=5)
lc_flat = lc_clean.flatten(window_length=401) # Detrending

# 3. Periodicity Search: Box-Least Squares (BLS)
period_grid = np.linspace(0.5, 5, 5000)
bls = lc_flat.to_periodogram(method="bls", period=period_grid)
best_period = bls.period_at_max_power
print(f"\n[DETECTED] Most probable orbital period: {best_period:.4f}")

# 4. Graphs Plotting (Fixed with 's' parameter instead of 'size')
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 8))

# Plot 1: Detrended Flux Time Series
lc_flat.scatter(ax=ax1, color='black', s=1, label='Detrended Flux')
ax1.set_title(f"{target_star} - Detrended Time Series Data")

# Plot 2: Phase Folded Light Curve (Transit Signal Visualized)
lc_folded = lc_flat.fold(period=best_period)
lc_folded.scatter(ax=ax2, color='blue', s=2, label=f'Folded at P = {best_period:.4f}')
ax2.set_title("Phase Folded Light Curve (Transit Signal Visualized)")

plt.tight_layout()
plt.show()
```

```
/usr/local/lib/python3.13/dist-packages/lightkurve/prf/__init__.py:7: Use
warnings.warn(
Searching data for Kepler-10...
/usr/local/lib/python3.13/dist-packages/lightkurve/search.py:421: Lightku
warnings.warn(
```

[DETECTED] Most probable orbital period: 0.8376 d

